

# Module 24.15: Logistic Regression Implementation & Evaluation

## 1. End-to-End Classification Pipeline (Heart Disease Dataset)

We execute an end-to-end binary classification model using the clinical `heart.csv` dataset to predict the presence of heart disease ( $y \in \{0, 1\}$ ).

```
+-----+
| 1. Ingestion: Inspect 303 medical records (13 clinical features, target 'target' |
| 2. Train-Test Split: 80% Train, 20% Test (stratified split) |
| 3. Feature Scaling: StandardScaler() on continuous predictors |
| 4. Model Training: LogisticRegression(max_iter=1000) |
| 5. Evaluation: Confusion Matrix, Precision, Recall, F1-Score, ROC-AUC |
+-----+
◀────────────────────────────────────────────────────────────────────────────────▶
```

## 2. The Confusion Matrix

Binary classification evaluation partitions predictions into a  $2 \times 2$  contingency matrix:

|                    | Actual Positive ( $y = 1$ )   | Actual Negative ( $y = 0$ )  |
|--------------------|-------------------------------|------------------------------|
| Predicted Positive | True Positive (TP)            | False Positive (FP) (Type I) |
| Predicted Negative | False Negative (FN) (Type II) | True Negative (TN)           |

◀────────────────────────────────────────────────────────────────────────────────▶

### 3. Core Classification Metrics

---

#### 1. Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Misleading on imbalanced datasets (e.g., in fraud detection where 99.9% of transactions are legitimate).

#### 2. Precision (Positive Predictive Value)

$$\text{Precision} = \frac{TP}{TP + FP}$$

- "Out of all patients predicted to have disease, how many actually had it?"
- Critical when False Positives are costly (e.g., Spam detection).

#### 3. Recall / Sensitivity (True Positive Rate)

$$\text{Recall} = \frac{TP}{TP + FN}$$

- "Out of all patients who genuinely had disease, how many did the model catch?"
- Critical when False Negatives are fatal (e.g., Cancer / Heart disease diagnosis).

#### 4. F1-Score (Harmonic Mean)

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Penalizes models that achieve high precision at the expense of terrible recall.

#### 5. ROC Curve & AUC Score

- **ROC Curve:** Plots True Positive Rate (Recall) vs False Positive Rate ( $FPR = \frac{FP}{FP+TN}$ ) across all classification thresholds  $\tau \in [0, 1]$ .
- **AUC (Area Under Curve):** Quantifies ranking discrimination capability ( $AUC = 1.0$  is perfect;  $0.5$  is random guessing).